

The geography of venture capital investments in the UK

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The role of venture capital in economic development increasingly is recognized as central to the development of an entrepreneurial economy. However, the supply of venture capital is not distributed evenly across the space economy. In the UK, evidence for the 1980s demonstrated that venture capital investments were highly concentrated in Greater London and the South East, reinforcing the existing patterns of regional concentration of economic activity. This paper reviews the regional distribution of venture capital investments in the UK in the 1990s, a period of massive growth in venture capital investment activity. It concludes that the regional concentration of venture capital investment has been considerably reduced since the 1980s. However, more detailed analysis of the data demonstrates that this shift towards a less unequal regional distribution has been driven by so-called 'merchant' venture capital – investments in large-scale management buy-outs and buy-ins which facilitate corporate restructuring through ownership change and often have adverse consequences for employment. 'Classic' venture capital – investments in young entrepreneurial companies with high growth potential – remains highly concentrated in London and the South East, and also in Scotland. This reflects both supply- and demand-side factors. The Government's new regional venture capital funds are unlikely to be effective in closing this regional finance gap. An alternative approach to intervention, in the context of the increasing globalization of venture capital investments, is to seek to attract venture capital money and expertise from elsewhere.

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Introduction

Since the late 1980s there has been a growing volume of work on the spatial dimensions of money, although, as Martin notes, 'we are still far from having a comprehensive synthesis encompassing the full range of financial geographies' (Martin 1999, 5). Nevertheless, studies of the spatial organization and operation of particular

financial institutions are highlighting how various institutions and markets that make up the financial system are shaping the wider system of production, employment and income of localities, regions and nations (Martin 1999). This paper is concerned with venture capital – a specialized form of industrial finance that provides equity capital both to young rapidly growing companies, often in technology sectors, that have the potential to

become significant global businesses and also to companies that are undergoing ownership transformations. Given these investment foci, its potential significance in contributing to uneven regional economic development is considerable.

Venture capital is a growing field of academic enquiry, but much of the research has been undertaken from a management perspective (see reviews by Wright and Robbie 1998; Mason and Harrison 1999a; Manigart and Sapienza 2000; Amit *et al.* 2000; Gompers and Lerner 2001a). This research has provided a sophisticated understanding of the 'technical' aspects of the investment process (Bygrave and Timmons 1992; Gompers and Lerner 1999 2001b). However, there has been much less progress in understanding the spatial characteristics of venture capital or its role in local and regional economic development (Mason and Harrison 1999b). This supports Martin's observation that 'the intersection between money and the local economy ... [is] thus far, the least explored aspect of the economic geography of money' (1999, 23–4).

The existing literature on the geography of venture capital largely relates to the 1980s, when the venture capital industry was in its early growth stage in the USA and in its infancy elsewhere (Mason 1987; Leinbach and Amrhein 1987; Florida and Kenney 1988; McNaughton and Green 1988; Martin 1989 1992; Mason and Harrison 1991 1999b; Florida and Smith 1993).

During the 1990s there was an enormous growth in venture capital in the UK.¹ The amounts raised by UK *independent* venture capital companies (largely from pension funds, banks and insurance companies) increased from less than £0.5 billion in the early 1990s to £9 billion in 2000: indeed, £26.9 billion was raised in the period 1997–2000, 79 per cent of the total amount raised over the decade (BVCA 2001a, Appendix 6, Table U).² The amounts invested increased from £1.2 billion in 1992 to £6.4 billion in 2000, a 433 per cent increase (BVCA 2001a, Appendix 6, Table A). However, preliminary figures for 2001 indicate that investment activity has fallen from this peak by some 25 per cent (BVCA 2002a), reflecting the cyclical nature of the industry.

Research in the 1980s has highlighted the extreme concentration of both office locations and funds invested and the existence of regional venture capital gaps, which Thompson (1989) defines as regions which are at a comparative disadvan-

tage in getting otherwise comparable projects funded. Given the expansion of the venture capital industry in the 1990s, the geographical distribution of investment assumes even greater significance in any debate about uneven development in the UK. This paper therefore goes beyond the largely static and highly aggregated analyses of venture capital investment activity in the 1980s to explore from a UK perspective whether the initial spatial concentration of venture capital investment activity has given way to some diffusion as the industry has grown and matured. Specifically, given that previous research has demonstrated a strong regional over-concentration of venture capital investment in the South East (Martin 1989 1992; Mason and Harrison 1991), the question which the paper seeks to answer is whether this pattern of investment activity has changed during the 1990s.

Thompson (1989) notes that there are two contrasting perspectives on this issue. A market-equilibrium perspective predicts that there will be a spatial diffusion of venture capital investment activity over time because competition for investment opportunities in the areas where venture capital initially concentrated will lead to declining returns and, in turn, force investors to seek untapped opportunities in other locations.³ This model, he argues, is too deterministic in relying on markets working to spatial equilibrium. He therefore offers a geographical model based on a more sophisticated picture of the relationships between institutional forms, investment processes and spatial outcomes. This model recognizes the spatial constraints on the investment behaviour of venture capitalists that arises from both their networks of informants and the need for spatial proximity to monitor investee companies. As a consequence, venture capital is inherently spatially restricted and limits the extent to which spatial diffusion occurs, and the nature of any spatial diffusion, with the possibility that some regions are never supplied.

The availability of venture capital is now on the policy agenda in most advanced countries. Venture capital is seen by policymakers in Europe and elsewhere as a key instrument to support economic growth and employment (Commission of the European Communities 1998; Harding 2000). In the case of the UK, increasing the availability of venture capital occupies a central position in the Labour Government's efforts to create a competitive, innovative, knowledge-driven economy:

venture capital needs to be available to innovative entrepreneurs who are

central . . . to promoting growth, increasing productivity and creating jobs . . . [They] sense opportunities to take risks in the face of uncertainty to open new markets, design products and develop innovative products. In the knowledge driven economy this process is even more crucial. (Department of Trade and Industry 1998, paras. 2.3, 2.19)

The Chancellor of the Exchequer, in his Pre-Budget Statement in November 1998, stressed that 'a strong venture capital industry supporting high tech, high risk investments is critical to the future of Britain'. More recently, the availability of venture capital has been linked to efforts to close the UK's productivity gap with its competitors (HM Treasury 2000 2001). In order to increase the availability of venture capital for smaller enterprises across the country, the Government is supporting the development of regional venture capital funds in every English region. These funds, which will benefit from Government and European Investment Fund investment, will operate as public-private partnerships. In early 2002, the Scottish Executive announced that a similar scheme for regional enterprise funds would be established in Scotland.

These efforts to promote venture capital can be seen in the context of the changing form, functioning and strategic direction of the state (MacLeod and Jones 1999; Painter 2000). Jessop (1994) has argued that the Keynesian welfare state is giving way to a Schumpeterian workfare regime. This involves a re-orientation of policy away from demand management, involving the state's direct management and sponsorship of social and economic projects, in favour of supply-side activities that are intended to encourage innovation in products, processes and markets to enhance international competitiveness. These interventions involve an increasing role for non-government organizations in both the formulation and implementation of public policy.

The structure of the paper is as follows. In the next two sections, we review the nature of venture capital and its role in the development of an entrepreneurial economy, and in the process of technology-based change in particular. This is followed by a review of the role of venture capital in regional development and, specifically, the implications of this for uneven development and

the perpetuation or reinforcement of regional inequalities. Subsequent sections of the paper review empirical evidence on the regional diffusion of venture capital investments in the UK in the 1990s. This highlights in particular the different geographies of venture capital support for new enterprise development (i.e. 'classic' venture capital) and for enterprise restructuring (i.e. 'merchant' venture capital). The paper concludes with a review and assessment of some policy implications arising from this analysis in the context of the establishment of regional enterprise funds in England and Scotland.

The nature of venture capital

The focus of this paper is restricted to the institutional venture capital industry and does not consider either the informal venture capital market, which comprises private individuals (business angels) (Mason and Harrison 1994 2000), or direct forms of investment by non-financial companies (corporate venture capital) (McNally 1994 1997; Gompers and Lerner 2001b, ch. 7). The institutional venture capital industry comprises three actors (Figure 1): *professional venture capital firms* who raise finance from *financial institutions* (e.g. banks, insurance companies and pension funds) and *other investors* (e.g. corporations, universities, wealthy families, government agencies) to invest in *businesses with high growth potential*. Venture capital firms have a specific expertise in reducing information-based market failures that arise from asymmetric information (Amit *et al.* 1998). By means of careful selection, monitoring, contracting and post-investment support, they are able to identify investment opportunities with the potential for high returns, support the development of their investee firms and secure an exit for the investment.

There are two types of venture capital company. First, *independent* venture capital firms, which are commonly organized as limited partnerships, although some quoted investment trusts exist, create one or more funds of a specified size, duration and investment focus, and then seek investors to invest in the fund. The fund is constructed as a separate legal entity, with the venture capital firm being the managing (or general) partners and the institutional investors in the fund being the limited partners. Generally, no single

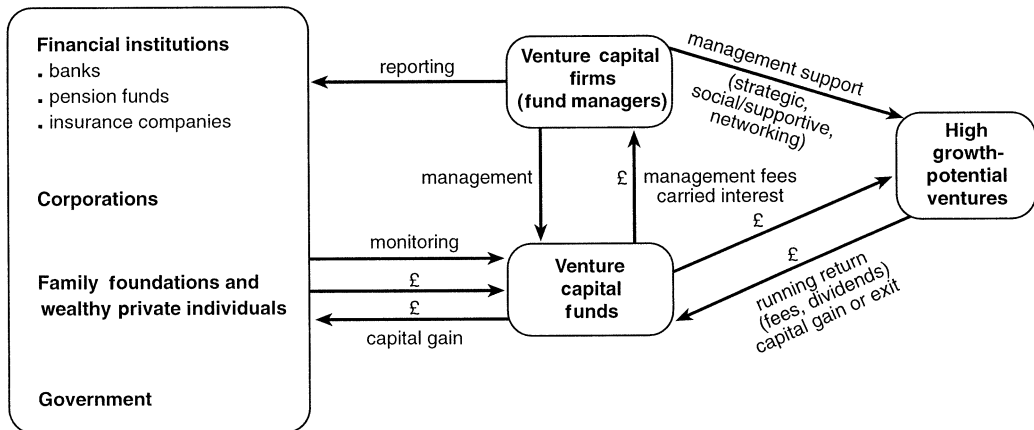


Figure 1 The venture capital investment process

investor or shareholder has a dominant position in the fund's ownership. The venture capital firm gets its return in two forms, partly as a fee for management services in the investment process, and partly as a share of the capital gains realized by the fund. Larger venture capital firms may manage three to five funds at any one time. Second, *captive* funds are the in-house venture capital subsidiaries of financial institutions (e.g. banks, insurance companies). Finance is obtained from their parent companies. Many captive funds have evolved into semi-captive funds, subsequently raising outside finance from other sources, which may be managed in parallel with in-house funds, in addition to the finance originally provided by their parent organizations.

In their analysis of the development of the venture capital industry in the USA in the 1980s, Bygrave and Timmons (1992) have suggested that the institutional venture capital industry actually comprises two distinct types of investor. First, *classic* venture capital is oriented towards the financing of companies at their seed, start-up and early growth stages. These funds raise their finance from patient investors, their managers have entrepreneurial experience and industry knowledge, whose skills are in company forming, building and harvesting, and who support their investee businesses through mentoring, networking, strategic insights and social support. Most classic venture capital involves the financing of technology-based firms. Second, *merchant* venture capital (or business development capital) funds raise their finance from institutional investors who have short-term invest-

ment horizons, are managed by MBA-trained professionals with backgrounds in investment banking or other financial or consulting organizations, who have strong analytical, financial engineering, deal-making and transaction crafting and closing skills but little company-building experience and so have little ability to add value to businesses. Merchant venture capital funds invest in later stage deals and in management buyouts (MBOs) and buy-ins (MBIs), that is, the sale by large companies of non-core subsidiaries and divisions to either incumbent or incoming management teams. MBOs and MBIs also play important roles in privatizations and in overcoming succession problems in family-owned businesses.⁴ Given the importance of recognizing the institutional geography of finance (Martin 1999), a key issue that emerges from this distinction between classic and merchant venture capital is the potential implication for regional development of different geographies of the two types of investment. This is an issue to which we return later in the paper.

The economic significance of venture capital

The importance of venture capital arises from the catalytic role that it plays in the entrepreneurial process by providing finance to companies seeking to exploit significant growth opportunities, usually as a result of technological innovation, but which are unable to fund this growth from internally

generated sources of finance and debt finance, and are too small to access public equity markets (Brophy 1997; Wright and Robbie 1998; Mason and Harrison 1999a; Gompers and Lerner 2001b; Norton 2001, 239–49). Young, fast-growing companies typically have negative cash flows in their early years of trading as a result of the need to make investments in R&D, product development and testing, purchase facilities and key equipment, recruit management, spend on raw materials, components, sales and marketing, and build up inventory. Indeed, the faster the firm grows, the more voracious is its appetite for cash (Bygrave and Timmons 1992; Binks and Ennew 1996). It is not uncommon for those firms which subsequently become successful to have a negative cash flow for two or three years and take more than five years to become profitable (Bygrave and Timmons 1992). The assets of such firms are likely to be intangible (e.g. patents, trademarks, human capital). These are difficult to value and have uncertain secondary markets and so are unacceptable as security by providers of debt finance (Gompers and Lerner 2001b), and any acceptable forms of personal collateral are likely to have been exhausted. For these reasons, banks will not finance such firms. However, venture capital firms understand the risks involved in financing such firms and possess the know how to overcome these risks and achieve commercial success. Moreover, they seek to add value to their investments by coaching the entrepreneur, providing strategic advice and assistance, attracting additional finance, managers and directors to the business and making the appropriate contacts with suppliers, customers and advisers. Without venture capital, such firms have a high probability of being aborted or failing in infancy (Bygrave and Timmons 1992; Gompers and Lerner 2001b).

Venture capital also plays a significant role in facilitating corporate restructuring and ownership change in family-owned businesses. Venture capital enables large companies to divest part of their operations – which may either be part of a planned strategic refocus or because of financial difficulties – by financing incumbent or incoming management to purchase the subsidiary or division through a management buy-out (MBO) or buy-in (MBI). MBOs and MBIs may also involve the profitable subsidiaries of failed groups. MBOs and MBIs are also a common vehicle for privatization. Venture capital also supports the succession of

ownership and management in family-owned companies where the only alternative may be sale to another business.

MBOs and MBIs create enhanced value by improved corporate governance arrangements involving a combination of management ownership (and associated incentives) and institutional involvement. First, ownership acts as an incentive to managers to strive for profits rather than personal sources of utility. Without an ownership stake managers are not entitled to the bulk of the profits generated through increased efficiency. MBO transactions typically involve incentive schemes; for example a ratchet mechanism might be put in place to allow management's final equity stake to reflect performance. Second, owner-managers will devote more effort to seeking out innovative projects, whereas without an ownership stake the risks of failure are not adequately compensated by a share of the benefits if the project is successful. Third, the venture capital investors are significant equity holders and so have strong motivation to monitor the performance of the management through board representation. Fourth, the financing structure, which involves a large amount of debt alongside the equity substantially reduces management's discretion by committing them to deliver on a performance plan agreed at the buyout (Green 1992; Thompson and Wright 1995; Zahra 1995). Thus, MBOs and MBIs are associated with improved operating performance (linked to improvements in working capital management, especially credit management, better cost control and asset disposals), modifications to existing product range, new product development, increased customer base and organizational changes (e.g. removing layers of hierarchy, quickening the pace of decision-making, speeding up the flow of information in the company; decreasing communication bottlenecks) (Kaplan 1989; Bull 1989; Smith 1990; Wright *et al.* 1992; Opler 1992; Smart and Waldfagel 1994; Zahra 1995; Robbie and Wright 1996; Robbie *et al.* 1999).⁵ The performance of MBIs is less strong than for MBOs and more extensive restructuring occurs. This is related to the less detailed knowledge of the business which the incoming management possess (Robbie and Wright 1996).

Evidence for the economic significance of venture capital is available at two levels. From a technology perspective, Kortum and Lerner (2000) argue, based on an analysis of US patent activity,

that venture capital drives innovation. Indeed, venture capital has played a

unique role in the formation and commercialisation of entirely new industries: personal computers, cellular communication, microcomputer software, biotechnology, and overnight delivery to name a few. (Bygrave and Timmons 1992, 2)

For example, von Burg and Kenney (2000) highlight the role played by venture capitalists in the emergence of the local area networking (LAN) industry. From an economic perspective, Gompers and Lerner note that

firms backed by venture capitalists grow more quickly and reach the public market sooner than similar non-venture firms ... [and] ... continue to outperform nonventure companies long after they go public. (Gompers and Lerner 2001b, 65–6)

Studies undertaken on behalf of the US, UK and European venture capital associations highlight the importance of venture capital in financing growing businesses. For example, the third study of the economic impact of venture capital undertaken on behalf of the British Venture Capital Association (BVCA 2001b) reports that in the five years from 1995/96: employment in venture capital-backed firms increased by 29 per cent per annum, against a national growth of just over 2 per cent; the annual sales growth of venture capital-backed companies was more than double that for FTSE 100 companies; exports grew by 27 per cent per annum compared with a national growth rate of 4.4 per cent and investment increased by 24 per cent per annum, compared with a national increase of 8.8 per cent. The majority of respondents stated that because of venture capital backing their employment, exports and R&D expenditure were all higher than would have been possible otherwise: 87 per cent of companies would not have existed, or would have grown less rapidly, without venture capital.⁶ From a regional perspective, venture capital is seen as playing a key role in the emergence of high tech clusters (Norton 2001; Mason *et al.* 2002).

Venture capital and regional development

There are four important issues associated with regional variations in the supply of venture capital: the location of venture capital investments, the inter-regional flows of capital (from investors in venture capital funds) which underlie the pattern of investment, the support for the development of

a local/regional technology infrastructure and the influence of the institutional structure of the venture capital industry (merchant vs classic venture capital).

Research in the 1980s indicated that the geographical distribution of venture capital investment activity was highly uneven, with the South East, along with parts of adjacent regions (notably East Anglia), attracting a disproportionate share of venture capital investments (Mason 1987; Martin 1989 1992; Mason and Harrison 1991). As a consequence, the economic benefits of venture capital were largely restricted to what was already the most economically dynamic part of the country (Martin 1988). The uneven geographical distribution of venture capital investments could therefore be identified as one of the more significant causes of the widening north–south divide in the 1980s (Martin 1988). As access to capital is a major constraint on the start-up and growth of firms (ACOST 1990; Storey and Tether 1998; Freel 1999; Oakey and Mukhtar 1999), the differential spatial access to the availability of risk capital may be expected to lead to spatial variations in entrepreneurial activity.

This uneven geography of venture capital activity has a further significance. As noted earlier, venture capital firms raise their investment capital from financial institutions, notably pension funds, insurance companies and banks which, in turn, are the repository of the savings of individuals throughout the country in the form of occupational pension contributions, insurance premiums, investment trusts/mutual funds and bank savings accounts. The effect of the uneven spatial distribution of venture capital investments is therefore to deplete venture capital-deficient regions – typically peripheral, economically depressed regions – of savings which are invested in a small number of economically buoyant geographical locations (Mason and Harrison 1989; Martin and Minns 1995). As such, the higher degree of regional concentration of venture capital investments compared with capital invested in venture capital funds contributes to the regionally destabilizing geography of money in advanced industrial economies (Clark 1999; Martin 1999).

Third, the importance of venture capital is not simply as a source of finance. As Florida and Kenney emphasize, venture capital firms are an integral part of a well-developed local technology infrastructure:

venture capitalists enhance such environments by acting as both catalyst and capitalist, providing the resources and the contacts to facilitate new business start-ups, spin-offs and expansions. Because they sit at the centre of extended networks linking financiers, entrepreneurs, corporate executives, head-hunters and consultants, venture capitalists have a propulsive effect on business formation. (Florida and Kenney 1988, 43–4)

This is important from an economic development perspective. In order to be successful, technological pioneers have to bundle together a number of inter-related competencies into a 'competence bloc' (Eliasson 1992 1996a 1996b):

the total infrastructure needed to *create* (innovation), *recognise* (venture capital provision), *diffuse* (spillovers) and successfully *exploit* (receiver competence) new ideas in clusters of firms. (Eliasson 1996a, 12)

The non-technical competencies of such a 'competence bloc' must be: first, entrepreneurial awareness, the realization of the marketability of a new product or technology; second, acquiring venture capital to finance the start-up firm; and third, managing the enterprise from start-up through expansion into maturity (Eliasson 1996a). A similar argument underlies Saxenian's (1994) discussion of venture capital in Silicon Valley: '[Silicon Valley venture capitalists] brought technical skills, operating experience and networks of industry contacts – as well as cash – to the ventures they funded' (Saxenian 1994, 39). Venture capital firms therefore need to be seen as key actors who contribute to the 'institutional thickness' of regions through their role in facilitating regional collective learning (Florida 1995; Asheim 1996; Morgan 1997; Simmie 1997; Keeble and Wilkinson 2000).

This leads on to the final point. This catalytic role in the creation of new companies and supporting their growth is limited to classic venture capital. Merchant venture capital, by contrast, de-emphasizes company-building skills, and instead seeks to add value by means of financial engineering skills which often involve corporate restructuring. Saxenian (1994) attributes the superiority of Silicon Valley over the North East to the different types of venture capital in each region. She quotes a former Wall Street executive who states that:

in New York, the money is generally managed by professional or promoter types. Out here [Silicon Valley], the venture capitalists tend to be entrepreneurs who created and built a company and then sold out. When problems occur with any of their investments,

they can step into the business and help out. (Saxenian 1994, 39)⁷

Venture capital trends in the UK: an overview

Institutional venture capital originated in the USA in the early post-war period, although rapid growth only occurred from the early 1980s following reductions in the rate of capital gains tax and new rules which explicitly allowed pension funds to invest in venture capital funds (Bygrave and Timmons 1992; Gompers 1994). Venture capital did not become established in the UK until the 1980s. Indeed, prior to 1979 there were just a handful of venture capital firms – of which the most significant was ICFC, the predecessor of 3i plc (Clark 1987; Coopey and Clark 1995). However, from the early 1980s an enormous expansion occurred in the number of venture capital firms. Key influences were government policy and changes in financial markets. The Conservative Government under Mrs Thatcher stimulated a more entrepreneurial economy, resulting in a higher quality and number of 'backable' managers and businesses.⁸ The Business Start-Up Scheme, later extended and renamed the Business Expansion Scheme, which provided tax incentives for private investors who invested in unquoted companies, led to the establishment of collective schemes to manage these investments. This gave a significant boost to the development of venture capital fund management experience. The introduction of the Unlisted Securities Market in 1981 made it easier for smaller firms to achieve a flotation and in the process provided venture capitalists with a potential exit route for their shareholdings. Another important influence was the economic turbulence of the 1980s, which created opportunities for MBOs and MBIs. In addition, a new form of entrepreneurial investor emerged in the 1980s, influenced in part by the US experience, which showed how investing in high tech companies was very profitable for the investor, which actively sought investment opportunities and played an important role in expanding the market in which it operated (Coopey and Clark 1995; BVCA 1998). The British Venture Capital Association was established in 1983 with 34 founding members and by the end of the decade its membership had reached more than 120. Its current membership comprises 155 full members⁹ (BVCA 2002b).

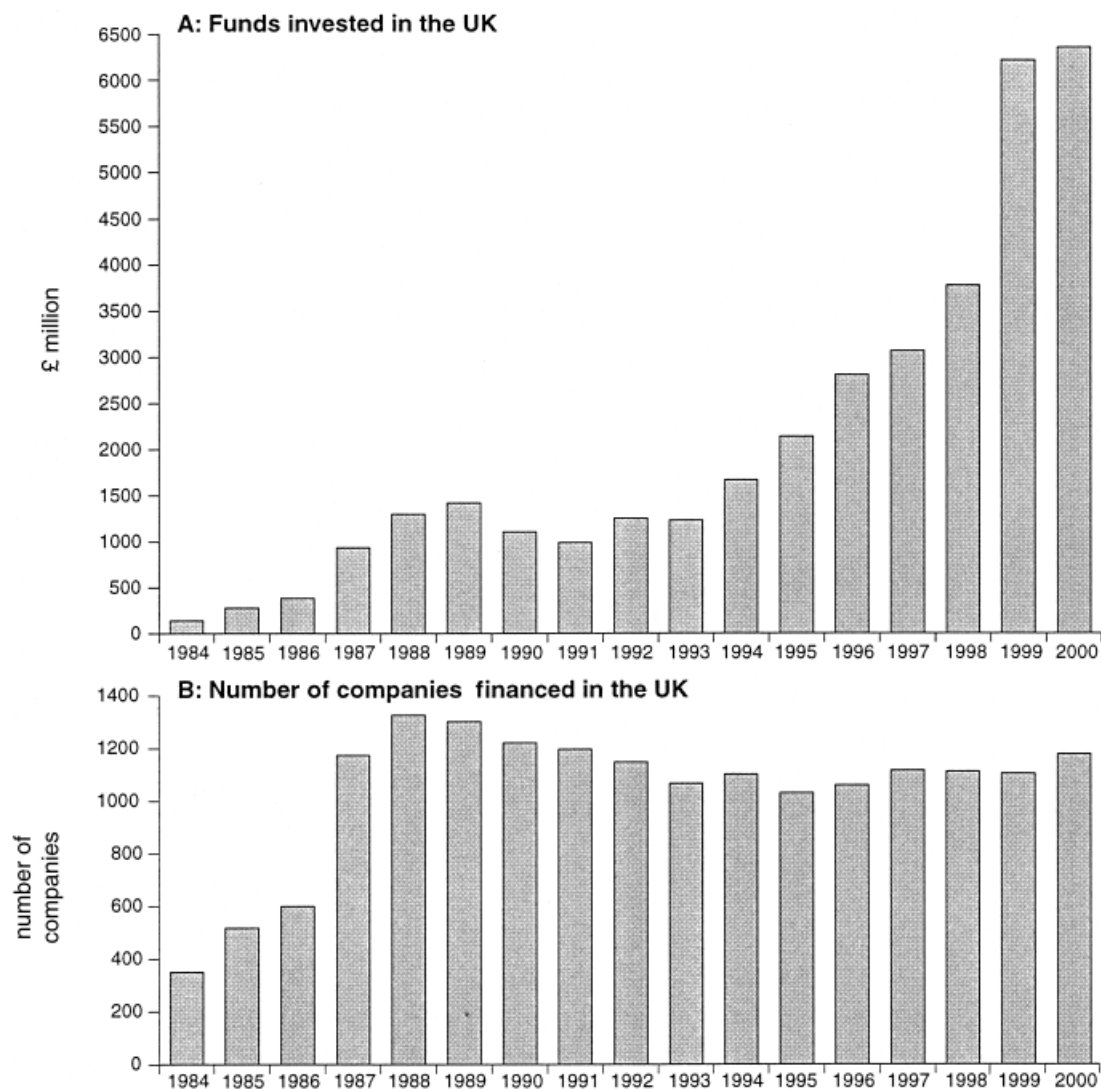


Figure 2 The growth of venture capital in the UK (a and b)

1. Figures not strictly comparable due to annual changes in membership
2. 1984–86 figures exclude 3i plc
3. Figures from 1992 onwards include secured debt where it is concurrent or alongside equity investment, or where it is rescue finance
Previously only unsecured debt as part of an equity/quasi-equity package was included

The total amount of funds invested grew rapidly from the early 1980s, peaking at £1.4 billion in 1989. This was followed by a period of decline and stagnation between 1989 and 1993 when various financial institutions' investors withdrew from the industry. Between 1993 and 2000 there was a rapid growth in the amount invested: the amount

invested in 2000 was nearly £6.4 billion, more than five times higher than the annual amounts invested in the early 1990s¹⁰ (Figure 2). However, despite the increasing amounts invested by venture capital firms, the number of investments has remained stable throughout the 1990s at just over 1000 (1182 in 2000), and is lower than the 1988 total (Figure 2),

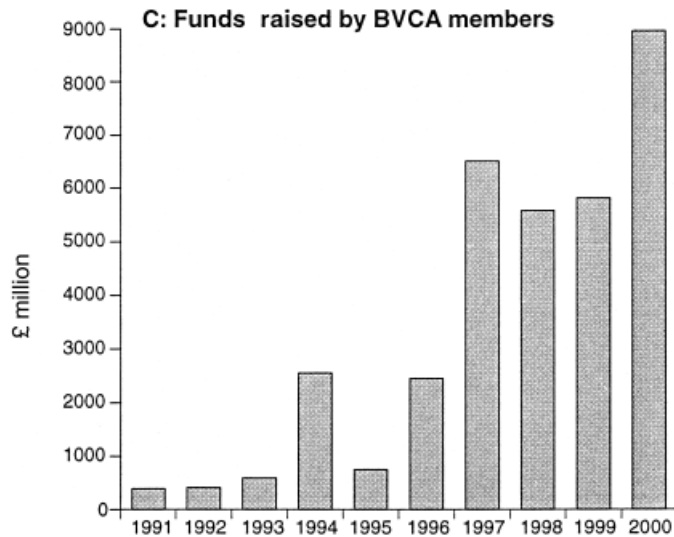


Figure 2 The growth of venture capital in the UK (c)

Note: Independent funds only

Source: BVCA (1998b) Appendix 6, Tables A and S

which reflects increasing deal sizes. UK venture capital activity experienced a significant downturn in 2001, in line with global trends. Although the number of companies financed increased by 10 per cent on the previous year, the amount invested declined to £4.8 billion, significantly down on the record levels in 1999 and 2000, but still higher than for 1998 and previous years¹¹ (BVCA 2002a).

These trends reflect the cyclical nature of venture capital investment that arises from fluctuations in the willingness of investors to provide money to venture capital firms that are driven by their return expectations. Successful exits in the early 1990s, which provided evidence of attractive returns (supported by the development of objective performance indicators), encouraged many investors to return and has attracted new investors. There has been a particularly significant flow of money from large North American investors seeking diversification from their domestic market (*Financial Times* 1999). These investors accounted for 41 per cent of the funds raised by independent UK venture capital firms in the period 1997–2000 inclusive (BVCA 2001a). This current downturn in venture capital investing in 2001 is the outcome of the overfunding in the late 1990s, the bursting of the technology 'bubble', and specifically the collapse

of various venture capital-backed dot-com businesses, the combined effect of which has been to drive down returns, and the end of the bull market which has prevented investors from achieving profitable exits from their investments.

This growth in venture capital investment activity during the 1990s has been accompanied by a significant change in the nature of venture capital investing. Since the late 1980s, the UK venture capital industry has become progressively more geared to investing in MBOs and MBIs. This reflects several factors (Wright *et al.* 1998, 151–3), notably the preference of institutional investors to invest in MBO funds rather than early stage funds, the superior returns achieved by MBO funds, and the shortage of classic venture capital skills. MBOs and MBIs represented 56 per cent of the amount invested in 2000, compared with only 21 per cent in 1984 (measured on a slightly different basis¹²) (Table I). However, this represents a decline compared with the late 1990s, when MBOs/MBIs were attracting over 70 per cent of venture capital investments by value. The increase in MBO/MBI investments has been at the expense of investments in seed capital, start-up and early stage ventures (Table II): although the actual amount invested in early stage ventures has increased substantially since the early 1990s and doubled between 1999

Table I Venture capital investments by stage

	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
A: amount invested (£m)																	
Early stage	38	50	86	120	130	215	128	58	82	69	76	85	131	159	288	347	703
Expansion	73	120	124	301	435	338	396	387	362	393	480	495	592	907	822	1156	2122
MBO/MBI	29	107	174	513	733	867	582	544	807	769	1112	1560	2083	2000	2665	4666	3546
Total amount invested	140	277	384	934	1298	1420	1106	989	1251	1231	1668	2140	2806	3066	3775	6169	6371
B: percentages																	
Early stage	27.1	18.1	22.4	12.8	10.0	15.1	11.6	5.9	6.6	5.6	4.6	4.0	4.7	5.2	7.6	5.6	11.0
Expansion	52.1	43.3	32.3	32.2	33.5	23.8	35.8	39.1	28.9	31.9	28.8	23.1	21.1	29.6	21.8	18.7	33.3
MBO/MBI	20.7	38.6	45.3	54.9	56.5	61.1	52.6	55.0	64.5	62.5	66.7	72.9	74.2	65.2	70.6	75.6	55.6

Note: Statistical discontinuities between 1986 and 1987 and 1991 and 1992: see notes to Figure 1 for details

Source: BVCA (2001a) Appendix 6, Table C

Table II Regional distribution of venture capital investments by value 1985–1991

	1985	1986	1987	1988	1989	1990	1991	1987–1991		Location quotient*
								Total	%	
A: Amount invested (£m)										
South East	171	246	529	640	689	674	513	3045	53.0	1.52
South West	18	18	67	53	93	34	72	319	5.6	0.58
East Anglia	16	20	24	38	39	51	27	179	3.1	0.75
West Midlands	19	11	40	119	82	64	93	398	6.9	0.81
East Midlands	6	10	71	86	103	70	47	377	6.6	0.97
Yorkshire–Humberside	7	13	68	80	58	57	74	337	5.9	0.78
North West	5	17	36	57	76	38	46	253	4.4	0.48
North	2	12	6	103	37	23	21	190	3.3	0.86
Scotland	18	26	50	81	217	81	73	502	8.7	1.20
Wales	7	11	42	27	19	12	17	117	2.0	0.39
Northern Ireland	–	–	1	14	7	2	6	30	0.5	0.16
Total	269	384	934	1298	1420	1106	989	5747	100	
B: Percentages										
South East	63.6	64.1	56.6	49.3	48.5	60.9	51.4			
South West	6.7	4.7	7.2	4.1	6.5	3.1	7.2			
East Anglia	5.9	5.2	2.6	2.9	2.7	4.6	2.7			
West Midlands	7.1	2.9	4.3	9.1	5.8	5.8	9.3			
East Midlands	2.2	2.6	7.6	6.6	7.3	6.3	4.7			
Yorkshire–Humberside	2.6	3.4	7.3	6.2	4.1	5.2	7.4			
North West	1.9	4.4	3.9	4.4	5.4	3.4	4.6			
North	0.7	3.1	0.6	7.9	2.6	2.1	2.1			
Scotland	6.7	6.8	5.4	6.2	15.3	7.3	7.3			
Wales	2.6	2.9	4.5	2.1	1.3	1.1	1.7			
Northern Ireland	–	–	0.1	1.1	0.5	0.2	0.6			
Total	100	100	100	100	100	100	100			

* Location quotient is calculated as $\%VC_i/\%bus_i$, where $\%VC_i$ = percentage share of venture capital investment in region i and $\%bus_i$ = percentage share of stock of VAT-registered businesses in region i

No regional breakdown for 1984. Statistical discontinuities between 1986 and 1987

Source: BVCA (2001a) Appendix 6, Table L

and 2000, it accounts for a smaller proportion of investment activity than in the 1980s. It is therefore the increasing significance of MBO/MBI investments that has been responsible for the rising average size of investment.

The geography of venture capital investments in the UK: 1985–1991

Previous studies of the geography of venture capital investment activity in the UK that examined the

regional distribution of investments in the 1980s noted that the pattern was highly uneven (Mason 1987; Martin 1989 1992; Mason and Harrison 1991). Martin (1989, 400) observed that 'over the course of the 1980s the ... venture capital industry has tended to concentrate investments in the south eastern and adjacent areas of the country'. The South East accounted for nearly two-thirds of the total amount invested in 1985 and 1986, almost twice its 'expected' share based on its share of the stock of VAT-registered businesses (33.4%). Although the South East's share of venture capital investments declined during the second half of the 1980s, it remained at over 50 per cent, still one-and-a-half times its 'expected' share. In 1985–6, East Anglia was the only other region with more than its 'expected' share of venture capital investments. In the 1987–91 period, the only other region with more than its 'expected' share of venture capital investments was Scotland. With this exception, venture capital shunned peripheral regions. The regions with the greatest deviation between their actual and 'expected' share of venture capital investments were (in ascending order) Northern Ireland, Wales, the North West, the South West, East Anglia and Yorkshire–Humberside (Table II).

Both Martin (1989) and Mason and Harrison (1991) have interpreted this uneven distribution as a reflection of both demand- and supply-side factors. On the *demand* side, the most suitable investment opportunities were found disproportionately in the South East: it had both the highest volume of new firms and the highest rate of new firm formation (Keeble and Walker 1994), the most fast-growing firms (Mason 1985) and the greatest concentration of technology-based firms (Keeble 1994). Firms in the South East were also more likely to seek external finance (Thwaites and Wynarczyk 1996). Those firms in peripheral regions which did seek venture capital were often handicapped by professional advisers who did not know how to 'package' funding proposals (Invest North West Working Party 1989). In addition, because of the pre-existing spatial division of labour in large firms, fewer opportunities have existed for MBOs and MBIs in peripheral regions, because their economies are dominated by production-oriented branch plants with a truncated range of management functions and which are integrated into intra-company flows of semi-finished goods (Phelps 1993).

The key *supply-side* influence on the uneven distribution of venture capital investments is the

concentration of venture capital firms in London, which in the 1980s contained nearly two-thirds of venture capital firms (Mason and Harrison 1991). This concentration arose from London's long-established position as the pre-eminent financial centre in the UK, containing various financial and related institutions and a diverse range of investment expertise and business consultancy services. It was therefore inevitable that the venture capital industry, a further specialist division of the finance industry, should develop there for reasons connected with the localizing effects of the spin-off process, the availability of professional labour and the existence of agglomeration economies (Martin 1989). Moreover, a significant number of the venture capital offices outside the South East were public sector creations (regional development agencies, local authority enterprise boards). As noted earlier, the only region, apart from the South East, with an over-concentration of venture capital investment in this period was Scotland, and this can be attributed at least in part to the presence of a strong indigenous financial sector which supported the development of a venture capital industry. Despite the expansion of a number of regional financial centres in the UK in the 1980s (Leyshon and Thrift 1989; Martin 1999), this was not reflected in any similar development of regionally based venture capital industries.

The location of venture capital funds influences the geographical distribution of investments because of the opportunity cost of the venture capitalist's time in searching for and monitoring investments (Gifford 1998). Distance impedes venture capitalists' access to information regarding both investment opportunities and the performance of investee businesses (Thompson 1989; Green 1991). Venture capital firms rely on information sharing with other venture capital firms, accountants, banks, lawyers, consultants, universities and a variety of other organizations to identify investment opportunities, organize investments and mobilize resources for its investee companies. Because of the intensive nature of this information flow, it tends to be personalized, informal and therefore localized (Florida and Kenney 1988). As a result, this biases investments by venture capital companies in favour of companies located relatively close to their office. The 'hands on' investment style of most venture capital firms, which involves venture capital firms in close contact with investee companies in order to contribute technical

skills, operating experience and networks, also discourages venture capitalists from investing in businesses that are not located within easy travelling distance because of the additional costs and greater difficulties of interacting on a face-to-face basis with the entrepreneur. Martin (1989 1992) noted that the propensity for London-based funds to invest in other regions declined with increasing distance from the South East. Thus, the lack of venture capital organizations across the UK and their concentration in London 'have undoubtedly influenced the relative availability of venture [capital] finance in the deficient regions' (Martin 1992, 164).

The geography of venture capital investments in the UK: 1992–2000

Aggregate patterns

The geographical distribution of venture capital investments in the 1990s has not been examined previously. Because of discontinuities in the statistics (see note 10) the most appropriate start date for this analysis is 1992, which also coincides with the beginnings of the current boom in venture capital investment activity. However, because of changes in regional boundaries in 1998, it is necessary to sub-divide the analysis into two separate periods: 1992–8 and 1998–2000 (the data for 1998 are available for both the old and new regions).¹³

It is clear that during the 1990s there has been a reduction in the previously highly uneven regional distribution of venture capital investment activity in the UK (Figures 3 and 4, Tables III and IV). The majority of regions have shared in the significant growth in venture capital investment activity since the mid 1990s, with particularly large increases in the West Midlands, East Midlands and North West. The proportion of venture capital invested in these regions during the 1990s has been greater than 'expected', given their proportion of the UK business stock, and is significantly higher than in the 1980s. Although the South East has also experienced an increase in the amount of venture capital invested, for most of the 1990s its rate of growth was below the national average. Thus, the South East's share of venture capital investments by value in the period 1992–8 fell to 42.5 per cent, significantly below the level of the 1980s, although still in excess of its share of the UK business stock (35%). In the 1998–2000 period, London had con-

siderably in excess of its 'fair share' of venture capital investments, whereas the South East had only marginally more investment than expected. In another reversal of 1980s trends, Scotland has attracted less than its 'expected' share of venture capital investment activity throughout the 1990s. Northern Ireland and Wales, along with the South West, and to a lesser extent the North, East Anglia/Eastern and Yorkshire–Humberside continue to be venture capital-deficient regions.

The erosion of London and the South East's excessive share of venture capital investment activity in the 1990s, along with significant increases in venture capital investments in the 'industrial heartland' regions of the East and West Midlands and North West, are consistent with the changing regional distribution of offices of venture capital firms (Table V). London, and to a lesser extent Cambridge, have captured much of the growth in the number of venture capital firms during the 1990s. However, there has also been a significant increase in the number of venture capital offices in the West Midlands and the North West and significant increases in the numbers of branch offices in Yorkshire–Humberside and Scotland (offset by falls in the numbers of venture capital firms with head offices in these regions). This has been driven primarily by the opening of regional branch offices since the late 1980s by London-based venture capital firms, and to a lesser extent by venture capital firms based in other cities, in order to develop local networks of professional intermediaries to better identify investment opportunities (Mason and Harrison 1991; Wright *et al.* 1994; Murray and Wright 1996). These regional offices mainly have an MBO/MBI and development capital investment focus (Gibbs 1991). The increase in the number of venture capital firms that have their head offices in the regions has been much less significant.

Merchant vs classic venture capital

Differentiating between MBO/MBI investments (merchant venture capital) and investments in firms at their early stage (classic venture capital) provides a further significant insight into the geography of venture capital investment in the UK. Nationally, merchant venture capital accounted for 66 per cent of all venture capital investments by value in the period 1997–2000, with classic venture capital accounting for 8 per cent. The figure for

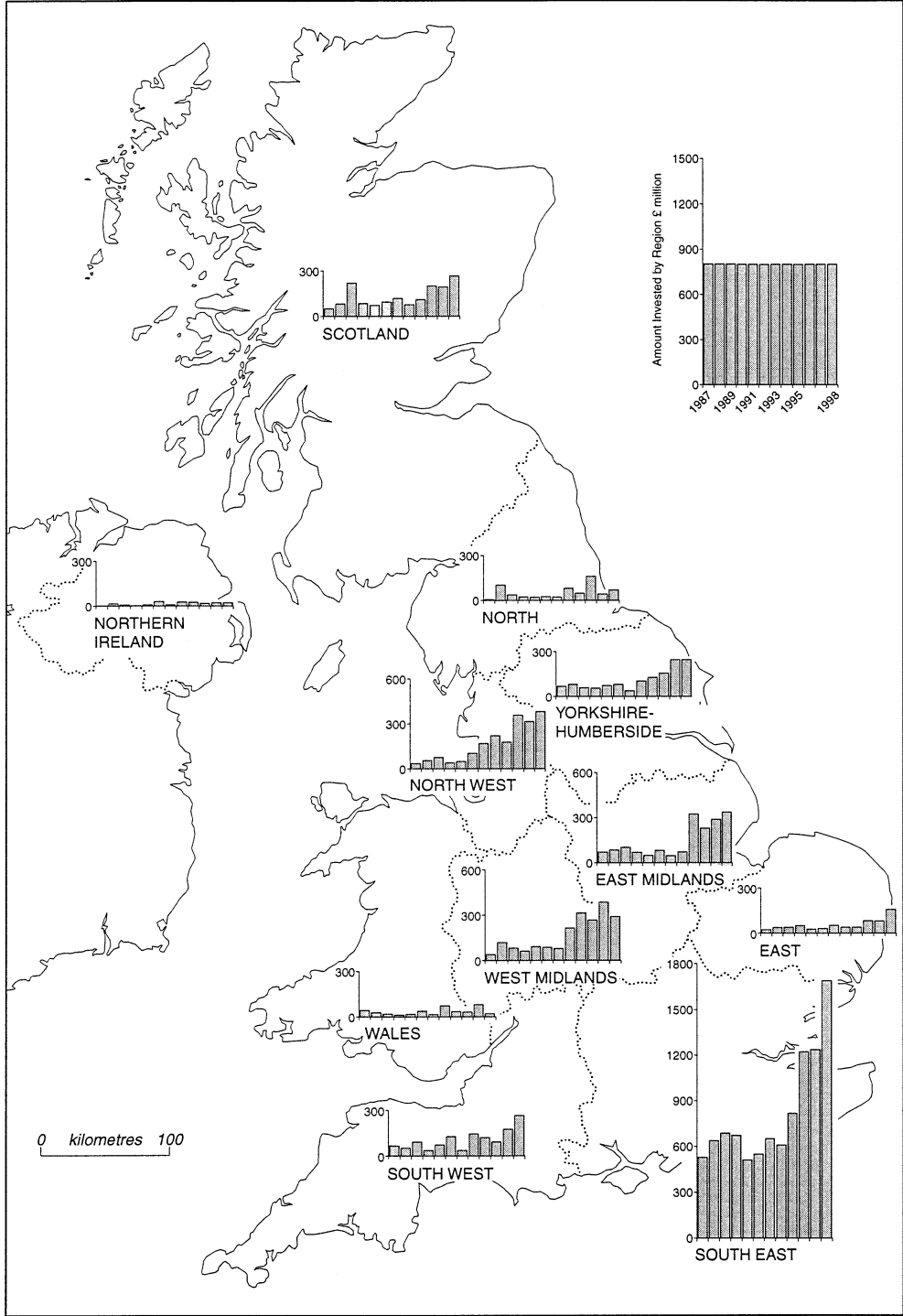


Figure 3 Regional distribution of venture capital investments by value, Standard Statistical Regions, 1987-1998

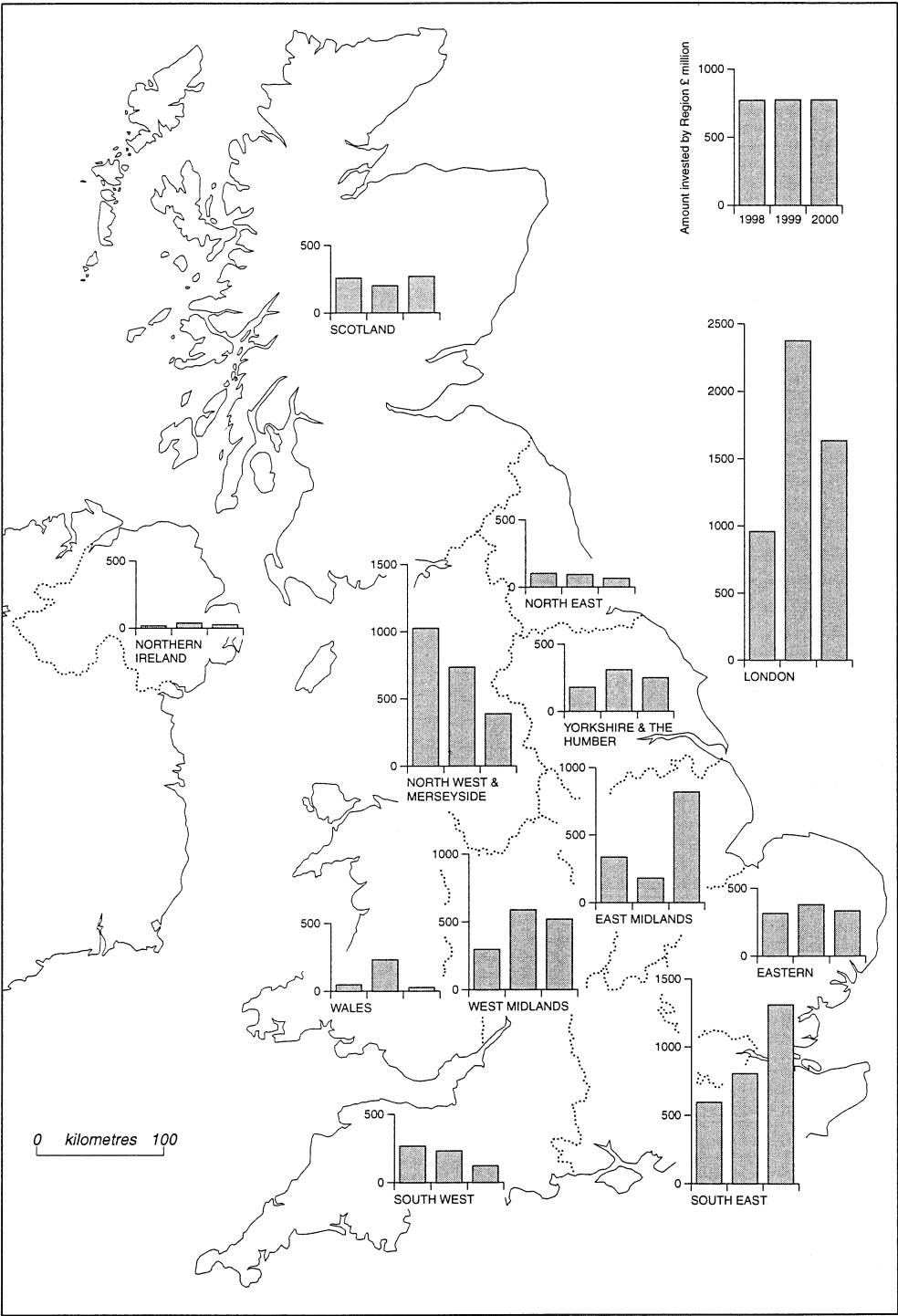


Figure 4 Regional distribution of venture capital investments by value, Government Office Regions, 1998–2000

Table III Regional distribution of venture capital investments by value 1992–1998

	1992		1993		1994		1995		1996		1997		1998		Cumulative: 1992–1998		Location quotient*
	£m	%	£m	%	£m	%	£m	%	£m	%	£m	%	£m	%	£m	%	
South East	550	44.0	650	52.8	612	36.7	819	38.3	1222	43.5	1228	40.1	1697	45.0	6778	42.5	1.21
South West	125	10.0	34	2.8	143	8.6	121	5.7	93	3.3	175	5.7	267	7.1	958	6.0	0.65
East Anglia	33	2.6	51	4.1	40	2.4	41	1.9	82	2.9	84	2.7	165	4.4	496	3.1	0.78
West Midlands	88	7.0	81	6.6	216	12.9	316	14.8	265	9.4	384	12.5	294	7.8	1664	10.3	1.21
East Midlands	83	6.6	46	3.7	73	4.4	322	15.0	228	8.1	291	9.5	331	8.8	1374	8.6	1.25
Yorkshire-Humberside	80	6.4	37	3.0	104	6.2	126	5.9	154	5.5	249	8.1	248	6.6	998	6.2	0.83
North West	106	8.5	168	13.6	225	13.5	180	8.4	355	12.7	317	10.3	386	10.2	1737	10.9	1.21
North	25	2.0	22	1.8	79	4.7	47	2.2	158	5.6	44	1.4	71	1.9	446	2.8	0.74
Scotland	96	7.7	119	9.7	77	4.6	109	5.1	200	7.1	196	6.4	275	7.3	1072	6.7	0.88
Wales	37	3.0	16	1.3	73	4.4	34	1.6	332	1.1	78	2.5	24	0.6	294	1.8	0.36
Northern Ireland	28	2.2	7	0.6	26	1.6	25	1.2	17	0.6	20	0.7	17	0.5	140	0.9	0.26
Total	1251	100	1231	100	1668	100	2140	100	2806	100	3066	100	3775	100			

* See notes for Table II for calculation of location quotient

Sources: (a) investment data: BVCA (2001a) Appendix 6, Table L; (b) business stock data: Small Business Service: *Business start-ups and closures: VAT registrations and de-registrations 1980–2000* (<http://www.dti.gov.uk/SME4/smehome.htm>) Accessed 28 May 2002

Table IV Investment by RDA region, cumulative totals 1998–2000

Region	Investments		Amount invested		Location quotients*	
	No.	%	£000	%	Investments	Amount invested
London	726	21.3	4971	30.5	1.28	1.84
South East	630	18.5	2699	16.5	1.20	1.07
South West	184	5.4	623	3.8	0.60	0.42
Eastern	284	8.3	1030	6.3	0.84	0.64
West Midlands	255	7.5	1405	8.6	0.91	1.05
East Midlands	188	5.5	1320	8.1	0.82	1.21
Yorkshire and the Humber	220	6.4	736	4.5	0.91	0.64
North West and Merseyside	310	9.1	2152	13.2	0.94	1.40
North East	114	3.3	253	1.6	1.32	0.61
Scotland	355	10.4	745	4.6	1.45	0.64
Wales	95	2.8	306	1.9	0.62	0.42
Northern Ireland	52	1.5	75	0.5	0.46	0.14
Total	3413	100	16 315	100		

Sources: as for Table III

merchant venture capital is significantly exceeded in two regions: the East Midlands (81%) and West Midlands (76%). Both of these regions, along with London, contain significantly more merchant venture capital than 'expected' (location quotients of 1.52, 1.28 and 1.78, respectively) (Table V). Classic venture capital is disproportionately concentrated in just three regions: London, the South East and Scotland (location quotients of 2.10, 1.43 and 1.19 under definition A in Table V). The North West also has a greater than expected proportion of classic venture capital if a wider definition is used (definition B in Table V). Indeed, Scotland and Northern Ireland (where the overall volume of venture capital investment is very low) are the only regions

where the amount of classic venture capital invested (definition B in Table V) exceeds that of merchant venture capital.

The aggregate analysis in the previous section highlighted the significant growth in venture capital investment activity in the industrial heartland regions, notably the East Midland, West Midlands and North West, and consequent decline in the over-concentration in London and the South East. The importance of this disaggregation is that it reveals that the growth of venture capital investment activity in these regions – especially in the East and West Midlands – during the 1990s has been largely in the form of investments in MBOs/ MBIs. As noted earlier, this type of venture capital

Table V Regional distribution of venture capital firm offices

	Main office				Change 1992–2002	Regional office				Change 1992–2002	Total				Change 1992–2002
	1992		2002			1992		2002			1992		2002		
	No.	%	No.	%		No.	%	No.	%		No.	%	No.	%	
Greater London	73	63.5	118	73.4	+45	7	10.1	5	6.8	-2	80	43.5	123	52.8	+43
Rest of the South East	8	7.0	9	5.6	+1	11	15.9	6	8.2	-5	19	10.3	15	6.4	-4
South West	1	0.9	0	-	-1	3	4.3	3	4.1	-	4	2.2	3	1.3	-1
East Anglia	3	2.6	6	3.4	+3	1	1.4	3	4.1	+2	4	2.2	9	3.9	+5
West Midlands	3	2.6	3	1.9	-	5	7.2	12	16.4	+7	8	4.3	15	6.4	+7
East Midlands	1	0.9	2	1.2	+1	2	2.9	2	2.7	-	3	1.6	4	1.7	+1
Yorkshire-Humberside	6	5.2	2	1.2	-4	7	10.1	11	15.1	+4	13	7.1	13	5.6	-
North West	3	2.6	5	3.1	+2	10	14.5	14	19.2	+4	13	7.1	19	8.1	+6
North	1	0.9	2	1.2	+1	3	4.3	2	2.7	-1	4	2.2	4	1.7	-
Scotland	11	9.6	7	4.4	-4	11	15.9	14	19.2	+3	22	12.0	21	9.0	-1
Wales	1	0.9	1	0.6	-	9	13.0	1	1.4	-8	10	5.4	2	0.8	-8
Northern Ireland	4	3.5	5	3.1	+1	3	4.3	0	-	-3	7	3.8	5	2.1	-2
Total	115	100	160	100	+45	72	100	73	100	+1	187	100	233	100	+46

Sources: calculated from CMBOR (1992) and BVCA (2002b)

investment activity is providing replacement equity to enable ownership change. Studies of the performance of MBOs in the immediate aftermath of the change in ownership – which were reviewed earlier – have highlighted significant improvements in working capital (especially credit) management, profits and productivity. However, cost cutting often makes a significant contribution to the gains in financial performance (Jensen 1993; Thompson and Wright 1995), with reductions in employment often following the buyout. Even though strategic actions taken in the aftermath of the buyouts are typically aimed at growing the business (e.g. new product development) (Wright *et al.* 1992; Zahra 1995; CMBOR 2000), job losses often result. US buyouts ‘are typically associated with initial shake-outs of employment’ and ‘when adjustments are made for industry factors . . . tend to recruit fewer employees than other firms in the same industry’ in the post buyout period (Wright *et al.* 1992, 64, 58). In the case of UK MBOs, the most comprehensive evidence (based on 182 responses) found that 24.7 per cent of firms had reduced employment on buyout and a further 17.6 per cent shed labour in the period after buyout. The loss of total employment in the sample was 6.3 per cent at the time of the buyout. By the time of the survey (approximately five years after the buyout), employment in the survey firms had recovered by 1.9 per cent, but was still 4.5 per cent below the pre-buyout level (Wright *et al.* 1992).¹⁴ Moreover, some of the increase in employment can

be attributed to acquisitions that some of the buyouts have made.¹⁵ A survey of MBOs in Europe (but biased towards the UK, which accounted for 66% of responses) in the period 1992–7 found evidence of plant closures (16%) and disinvestments (18%) in a significant minority of cases. Job losses following the buy-out were reported in just over one-quarter of cases and employment remained the same in a further 13 per cent of cases (CMBOR 2000).

Moreover, the longer-term benefits of buyouts have been questioned. It has been argued that buyouts are a ‘shock therapy’ whereby managers are incentivized to undertake one-off restructuring that produces short-term benefits, particularly in terms of financial performance, which are not sustainable in the future without further change (Rappaport 1990). This is attributed to limitations in MBOs as an organizational form, notably the limited strategic flexibility imposed by high debt levels, the desire of institutional investors to cash out and the absence of daily stock prices as an objective measure of corporate value (Kaplan 1991).¹⁶

The geographical distribution of merchant venture capital investment activity is consistent with both supply side trends and also the availability of buy-out and buy-in opportunities. On the supply side, as noted above, London-based venture capital firms have opened offices in provincial cities such as Birmingham, Manchester and Leeds to improve their deal flow by establishing direct access to

Table VI Regional distribution of classic and merchant venture capital 1998–2000

	Classic VC Definition A:		Classic VC Definition B:		Merchant VC		Location quotients		
	Early stage		Early stage and expansion financing		MBOs/MBIs		Classic VC		Merchant VC
	£m	%	£m	%	£m	%	Definition A: (early stage)	Definition B: (early stage and expansion)	MBOs/MBIs
London	329	22.0	1063	16.3	2193	17.0	1.43	1.06	1.10
South East	522	34.9	1680	25.8	3802	29.5	2.10	1.55	1.78
South West	67	4.5	272	4.2	526	4.1	0.50	0.47	0.45
East of England	111	7.4	413	6.3	860	6.7	0.76	0.64	0.68
West Midlands	62	4.1	438	6.7	1351	10.5	0.50	0.82	1.28
East Midlands	45	3.0	300	4.6	1311	10.1	0.44	0.69	1.52
Yorkshire and the Humber	76	5.1	299	4.6	685	5.3	0.72	0.65	0.75
North West and Merseyside	103	6.9	155	19.3	1231	9.6	0.71	1.99	0.99
North East	15	1.0	83	1.3	198	1.5	0.40	0.52	0.60
Scotland	129	8.6	526	8.1	415	3.2	1.19	1.12	0.44
Wales	14	0.9	116	1.8	268	2.1	0.20	0.40	0.47
Northern Ireland	24	1.6	59	0.9	37	0.3	0.48	0.27	0.01
Total	1497	100	6504	100	12 877	100			

Source: BVCA (2001a) Appendix 6, Table O

regionally based intermediaries such as accountants and legal firms (Wright *et al.* 1994; Murray and Wright 1996). The investment focus of these offices is mainly MBOs/MBIs. On the demand side, MBOs and MBIs reflect the spatial division of labour in large firms, with the greatest opportunities in regions with head office locations, autonomous subsidiaries and establishments that are self-contained in terms of production, management, sales and marketing and R&D, and lowest in regions where the ownership structure is weighted towards branch plants with a truncated range of managerial and technical functions. Rural regions, such as Eastern England and the South West, also have few opportunities for MBOs and MBIs because of their lack of large firms. In such regions, the main opportunities for MBOs and MBIs are confined to family-owned businesses with succession difficulties.

Classic venture capital, on the other hand, involves the creation and nurturing of entrepreneurial companies that are likely to exhibit innovative characteristics and create significant numbers of new jobs.¹⁷ This type of venture capital remains disproportionately concentrated in London and the South East, as well as in Scotland. Indeed, the London and South East regions attracted more than half of all the early stage capital invested in the period 1997–2000 (Table V).¹⁸

The geographical distribution of classic venture capital is consistent with the high rate of new firm formation and the concentration of technology-based firms and high growth entrepreneurial firms in London and the South East.¹⁹ However, supply-side factors are also significant. Classic venture capital funds are heavily involved with their investee companies in both a monitoring and advisory capacity because the management team in most early stage ventures is inexperienced and there is greater uncertainty associated with the technology, markets and competitive situation (Gupta and Sapienza 1992). Venture capital firms have less need to closely monitor their MBO/MBI investments, or to become actively involved, because these firms have an established management team, established products or services and established markets. Thus, the market for early stage venture capital investments is likely to be highly localized, whereas distance may be a somewhat less important consideration, at least from a monitoring perspective, in the case of MBO/MBI investments. The locations of investors specializing in making classic venture capital investments (Table VII) is consistent with the over-concentration of classic venture capital investments (especially early stage) in London and the South East, while Scotland's greater than expected share of classic venture capital investment activity is related to the investment activity of the venture capital division of

Table VII Regional distribution of 'classic' venture capital funds: 2002

	HQ	Branch	Total	%
Greater London	45	1	46	45.1
South East	8	1	9	8.8
South West	0	1	1	1.0
East of England	5	3	8	7.8
West Midlands	1	1	2	2.0
East Midlands	3	2	5	4.9
Yorkshire and the Humber	2	2	4	3.9
North West and Merseyside	4	3	7	6.9
North East	2	2	4	3.9
Wales	1	1	2	2.0
Scotland	4	6	10	9.8
Northern Ireland	4	0	4	3.9
Total	79	23	102	

Note: Classic venture capital funds are defined as funds whose average investment is less than £2 million and maximum investment is less than £5 million

Source: calculated from BVCA (2002b)

Scottish Enterprise²⁰ (which has had a significant catalytic effect on the development of an indigenous regional venture capital industry through both syndication and demonstration effects) (Hood 2000), the strong regional presence of 3i plc (with offices in Glasgow and Aberdeen) and a strong indigenous financial sector.

Conclusion

Summary

Venture capital plays a key role in the entrepreneurial process and in facilitating corporate restructuring and ownership change, including the re-indigenization of regional corporate ownership structures. Consequently, it is of considerable significance in shaping the pattern of regional economic development. In the 1980s, the UK's South East region attracted a disproportionate share of venture capital investments, whereas, with the sole exception of Scotland, the rest of the country attracted less than their 'fair share'. This paper has sought to answer the question: have venture capital investments become more evenly distributed across the regions of the UK during the 1990s, a period in which there has been a substantial growth in venture capital investment activity?

Our answer to this question is a qualified 'yes': venture capital investments have become more

evenly distributed. The South East's share of venture capital investments by value was lower in the 1992–8 period than in period from 1985 to 1991, although it was still higher than 'expected' when compared with the region's share of the total stock of VAT-registered businesses. In the period 1998–2000, based on the new Regional Development Areas, London continued to attract significantly more than its 'fair share' of venture capital investments, but the South East's share was only marginally greater than 'expected'. The decline in London and the South East's share of venture capital investments arises from the significant growth of venture capital investment activity in other UK regions during the 1990s, notably the West Midlands, East Midlands and North West. Indeed, these regions, along with London and the South East, have more than 'expected' shares of venture capital investments (i.e. location quotients greater than one). Northern Ireland and Wales, along with the South West, continue to have considerably less than expected shares of venture capital investment activity. Other regions with location quotients of less than one are the North/North East, East Anglia/East of England, Yorkshire–Humberside and Scotland.

The significant qualification is that the growth of venture capital investment activity in the regions, particularly in the East and West Midlands, has been of the merchant venture capital type, which involves providing replacement equity to enable incumbent or incoming management teams to buy out the existing owners of businesses, and is often associated with job losses arising from the restructuring that the new owners undertake. In contrast, classic venture capital, which is involved in nurturing innovative, fast-growing entrepreneurial businesses that are likely to be creating jobs, remains significantly over-concentrated in London and the South East, and also in Scotland and, using a wider definition of classic venture capital, the North West.

Policy implications

The evidence in this paper provides *a priori* support for the Labour Government's initiative to establish regional venture capital funds in each English region (Scotland, Wales and Northern Ireland have their own mechanisms for venture capital support) that were first announced in the 1998 Competitiveness White Paper (Department of

Trade and Industry 1998). These funds will specialize in making investments in amounts of less than £500 000: they are permitted to invest up to £250 000 as an initial investment in a company and may subsequently invest up to an additional £250 000 in follow-on funding. They will also be required to observe certain conditions relating to the types and mix of investment instruments (primarily in the form of equity) and certain industry sectors are excluded (Department of Trade and Industry 1999). The funds will operate as public-private partnerships. The funds will be commercially focused and managed by private fund managers, but up to half of the investment in each fund is being provided by Government and the European Investment Fund, with the Government's contribution provided on a subordinated basis²¹ (in order to raise the potential returns to private investors). Two funds were launched in late 2001, with the remainder being launched during 2002.

However, there are several grounds for questioning the likely effectiveness of this initiative (see Mason and Harrison 2003, for further elaboration). First, will the funds – particularly those outside London and the South East – be able to attract private investors? UK pension funds have exhibited a reluctance to invest in venture capital funds (Myners 2001), while local authority pension funds – which might be expected to have a commitment to local and regional economic development, have limited involvement in investing in either local or national venture capital funds (Doran and Bannock 2000). The Government accepts that it will need to provide incentives to encourage investors to switch part of their investment portfolios to this type of investment. However, the experience of the EU Seed Capital Funds suggests that financial institutions will be reluctant to invest in regional venture capital funds even if incentives are offered (Murray 1994a 1998). In practice, it appears that the private sector has been more willing to invest in funds in the 'south' (Edge 2001).

Second, are there sufficient fund managers who are experienced in making classic venture capital investments? Providing finance to new and young entrepreneurial companies is just one part of what classic venture capitalists do. Their effectiveness is ultimately contingent upon the specialist value added inputs and support services that venture capitalists can provide either directly or through their networks (Florida and Kenney 1988).

However, such classic venture capital skills are in short supply. Moreover, because of their low level of classic venture capital investment activity many regions lack the appropriate infrastructure of individuals and organizations (e.g. accountancy and law practices) that are sufficiently familiar with venture capital to support the fund managers.

Third, the ability of regional venture capital funds focusing on equity gap investments to achieve longer-term financial viability is likely to be problematic. This is, in part, because there is a difficulty in creating funds of a viable size (Murray 1994a 1998). It also reflects the fixed management costs involved in operating a venture capital fund, which have to be paid out of the capital that the fund has raised from its investors: the smaller the fund the greater the proportion of the capital raised that is accounted for by these costs.²² The small size of early stage funds creates a number of sources of vulnerability (Murray 1999):

- Their small size means that they can only make a limited number of investments. The narrow spread of investments increases risk because of limited diversification.
- The fund may be unable to provide follow-on finance to those firms in their portfolios that require a second round of investment (probably successful and consequently cash consuming investee companies), either because they have run out of money or because of constraints on the proportion of the fund that is permitted to be invested in one company. In such circumstances the fund will need to invite a larger fund as co-investor. However, the fund will be in a weak bargaining position and may have to accept a low equity revaluation and consequent dilution of its original investment (Murray 1994b).
- There is a strong probability that the funds will run out of money before they can 'harvest' their existing investment portfolio, but without an investment track-record are unlikely to be able to raise additional finance from private investors.

Furthermore, and related to the problem of the effects of small fund size upon financial viability, early stage funds have historically produced relatively poorer and less secure returns (i.e. higher relative dispersion of returns around the mean)

compared with those generated by later stage funds, and large MBO funds in particular (Burgyl 2000). This is in contrast to the USA, where early stage funds have demonstrated a superior performance to later stage funds. This may reflect the inability of UK investors to manage the particular and specific needs of early stage investments (Murray 1999) and, in turn, may be interpreted as being a consequence of the lack of classic venture capital skills in the UK.

Fourth, the initiative is based on the assumption that the financing gap reflects supply side problems. The possible role of demand-side deficiencies is not considered. However, both business angels and regional seed capital funds consistently report that they are unable to find sufficient investment opportunities (Harding 1999; Mason and Harrison 1999c 2001 2002). Demand constraints take three forms: aversion to using venture capital as a tool for business development; the restricted supply of sufficient numbers of viable, high growth potential businesses; and a lack of 'investment readiness' amongst those businesses who seek venture capital (Mason and Harrison 2001). There is no evidence to indicate the geographical dimension of such demand constraints, but it seems reasonable to speculate that they will be most acute in those regions that lack indigenous sources of classic venture capital. The clear implication is that simply creating pools of venture capital will not, of itself, lead to an increase in the amount of early stage venture capital that is invested. Indeed, the likely effect of creating an additional supply of early stage venture capital, operating on a less than fully commercial basis as a result of government financial support, in a situation in which there is a restricted supply of viable, high potential businesses, will be to create distortions in the market which over the longer term could drive out existing private sector venture investors. Where private sector venture capital funds are in competition with subsidized private sector funds for the same set of investment opportunities, they will be at a significant disadvantage both in terms of their higher costs and higher rate of return requirements to satisfy the investment targets of their investing institutions. Government-supported funds will therefore be able to offer better terms to potential investee businesses. One possible consequence is that private funds will withdraw from the market. Alternatively, if private venture capital companies choose to compete with government-supported

funds, the effect will be to drive down their rates of return, which, over the longer term, will discourage institutions and private individuals from providing further investment funds. In this situation private sector venture capital firms will be forced to withdraw from the market because they are unable to raise new funds (Riding and Orser 1997).²³

Thus, there must be serious doubts about the likely effectiveness of the new regional venture capital funds in achieving a significant equalization in the availability of classic venture capital across the UK regions. An alternative approach – proposed by Scottish Enterprise in its recent *Global Connections* strategy – is for regions to seek to import venture capital expertise and money from other countries and regions. Deficiencies in our understanding of the geographical operation of venture capital firms, which arises from the absence of any explicit geographical theory of the venture capital market (Martin *et al.* 2002), limits our ability to assess what impact this proposal might have. It clearly conflicts with existing geographical studies of venture capital activity which emphasize the localized nature of the investment process as a means of reducing investor risk (e.g. Florida and Kenney 1988; Green 1991; Martin *et al.* 2002; Zook 2002; Powell *et al.* 2002). Although long distance investing occurs where deals are syndicated between several venture capital firms, the presence of a local venture capitalist to lead the deal is generally essential (Florida and Kenney 1988). As our analysis has shown, in many UK regions there is a deficiency of local investors, especially ones with sufficient credibility, to play the lead role. However, the venture capital industry is in the process of evolving 'from a small, artisan business to a professional service organization' (Gompers and Lerner 2001b, 17). This is involving changes in organizational structure and strategies, one significant dimension of which is the change in the spatial organization and operation of the industry. International and global venture capital firms have emerged that raise pan-national funds to invest in several countries, both through syndication with local investors and by having a presence in several countries. Some of these venture capital firms specialize in large, later stage deals, while others are niche players who specialize in particular 'technology spaces' and locate their offices in globally significant technology clusters. For example, 32 per cent of the full

members of the British Venture Capital Association (many of which are affiliates of foreign firms) have at least one office in another country (excluding Ireland) and 7 per cent can be considered truly global, with at least five offices and operating on at least three continents. Paralleling this globalization of venture capital firms has been an increase in investments by financial institutions in overseas venture capital firms. Baygan and Freudenberg (2000) have highlighted the significant international flows of venture capital, while studies by Mason *et al.* (2002) and Leng (2002) have drawn attention to the important role of international flows of venture capital in the development of high technology industry in Ottawa and Silicon Valley.

Geographers have been slow to examine this transformation of the venture capital industry, understand the nature of this new economic geography of money or explore the consequences for local, regional and national economic development. This topic, which impinges on several issues at the forefront of contemporary economic geography – including economic globalization, geographies of finance and high-tech cluster dynamics – should therefore occupy a prominent position in the economic geography research agenda.

Notes

- 1 It should be noted that the growth of venture capital in the UK is part of a global trend, with North America, Europe and Asia all experiencing significant expansion in venture capital activity during the 1990s (Mason and Harrison 1999a).
- 2 These amounts exclude the money raised by captive venture capital funds – i.e. the in-house venture capital subsidiaries of financial institutions (e.g. banks) – and by semi-captive funds – i.e. captive funds which invest on behalf of their parent company as well as raising finance from independent sources.
- 3 Martin (1989) notes that by the late 1980s various observers were suggesting that the venture capital industry was entering a new phase of its development involving the creation of local venture capital funds in the regions and consequent reduction in the dependence upon London-based organizations for finance.
- 4 The term private equity is now being used to cover both classic and merchant capital, with the term venture capital used in a more restrictive way to cover classic venture capital.
- 5 The spatial outcomes of MBOs and MBIs have largely been ignored. The only study is by Wrigley (1999) in a unique analysis of how the capital structure transformation of a firm, following a MBO, has spatial outcomes in a study of the US food retail industry.
- 6 There is obviously a selection bias in these figures as venture capitalists search out and invest in high growth potential businesses.
- 7 This distinction is similar to that drawn between classic and merchant venture capital. However, Saxenian's argument should not be extrapolated to suggest that all Silicon Valley venture capital is classic venture capital and all East Coast venture capital is merchant capital (see Bygrave and Timmons 1992).
- 8 The BVCA's first executive director recalls that 'it was the day after Mrs Thatcher's government had cut the top rate of income tax in 1979 that Ronald Cohen, Sir David Cooksey, Gordon Dear and Tony Lorenz decided that the environment was right to set up the British Venture Capital Association' (BVCA 1998, 7).
- 9 The BVCA also has 143 associate members. These are professional firms including accountants, lawyers and consultants advising on private equity transactions or organizations that invest in private equity as a minor part of their overall activities.
- 10 Since 1992 the statistics on venture capital investment activity published by the British Venture Capital Association have included secured debt where it is concurrent alongside equity investment or where it is rescue finance. Prior to 1992, only unsecured debt as part of an equity/quasi-equity package was included (BVCA 2001a).
- 11 The reduction in the average size of deal in 2001 can be attributed in part to the decline in valuations following the end of the bull market.
- 12 See note 10. Because MBO/MBI transactions generally involve substantial amounts of debt and quasi-equity, one effect of the change has been to inflate this type of investment.
- 13 This change involved a switch from Government Office Regions (GOR) to Regional Development Agency (RDA) regions. However, only four GORs are affected: (i) three new RDA regions – South East, London and East of England – have been created out of the South East and East Anglia GORs; (ii) the North West RDA includes Cumbria, whereas the North West GOR did not; Cumbria was previously part of the Northern GOR. The North East RDA is the former Northern GOR without Cumbria.
- 14 However, the extent of job losses following MBOs may be related to the time-period. Robbie and Wright (1996) note that the scale of job losses was much higher amongst buyouts that took place in the early 1980s than those that occurred later in the 1980s. This reflects changes in the source of buyouts, with fewer buyouts in the mid and late 1980s being associated with companies in receivership and in financial difficulties. More recent research from the Centre

for Management Buyout Research (University of Nottingham) notes that the rationale for MBOs has shifted from cost reduction and strategic reorientation in mature sectors to creating value in technology sectors through product development and innovation (Robbie *et al.* 1999; Wright *et al.* 2001).

- 15 Equally, some of the employment loss can be attributed to divestments.
- 16 The evidence is not clear cut on this point. One UK study found that 21.4 per cent of buyouts in the period 1981–90 had undergone further ownership change – flotation, sale to a third party, secondary buy-out/buy-in (Wright *et al.* 1995).
- 17 The second BVCA impact study indicated that non-MBO venture capital-backed companies increased their employment by 33 per cent per annum between 1993/94 and 1997/98 compared with 15 per cent by MBOs (PriceWaterhouseCoopers 1999). Moreover, employment growth in MBOs is more likely to derive from acquisitions than in the case of non-MBOs. An extreme example of this is Capita Group plc, an MBO from CIPFA, which provides IT outsourcing services mainly for local authorities and other public sector organizations: 60 per cent of its present workforce of 7000 were transferred from the public sector as part of long-term outsourcing contracts.
- 18 In view of the dynamic high-tech cluster in the Cambridge region, which has been largely created by local entrepreneurial activity (Keeble *et al.* 1999), plus the increase in the number of Cambridge-based venture capital firms (Table V), it is surprising that the East of England does not have an over-concentration of classic venture capital. In the absence of statistics on investment activity below the regional scale, it can only be speculated that the remainder of the region has attracted only limited amounts of venture capital.
- 19 For example, 62 per cent of *Fast Track 100* companies in the period 1997–2001 inclusive are located in Greater London and the South East regions (<http://www.fasttrack100.co.uk>) Accessed 30 May 2002.
- 20 This division has now been spun-out of Scottish Enterprise as Scottish Equity Partners and operates on normal commercial terms as an independent venture capital company (Hood 2000).
- 21 This is being done in two ways: (i) if a fund makes a loss private investors will be repaid before Government; (ii) the Government's return is capped, unless the fund starts to make 'supernormal' profits (Campbell 2001).
- 22 Murray's (1994a 1998) evaluation of the EU's Seed Capital Fund Scheme estimated that because of their small size, the funds would exhaust their capital in between 9 and 11 years, even if no investments were made, purely as a result of their operating costs.
- 23 This issue has belatedly been recognized by Government (HM Treasury/Small Business Service

2001) and a number of local 'investment ready demonstration projects' are to be piloted with a view to a future national programme.

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